

A Systematic Literature Review of Cybersecurity Awareness and Strategy in Rural-Based Universities

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Abstract:

The number of cyberattacks against higher education organizations has dramatically increased in recent years, especially among the less-resourced rural universities where awareness and preparedness are sometimes non-existent. Often, these vulnerabilities can be attributed to human factors and lack of formal training or institutional strategies, if they exist. This paper provides a Systematic Literature Review (SLR) of Cybersecurity Awareness and Strategy in Rural based Universities. The study focused on literature from 2020-2025 analyzing the relationship between cybersecurity awareness and strategic cybersecurity in rural university environments. An exploration of SLR identified key themes such as awareness campaigns, organizational culture, digital literacy, institutional governance, and policy integration. The findings indicated that while awareness does improve the individual and organizational position related to cybersecurity, rural-based universities are faced with ongoing challenges relating to enforced policy, infrastructure, and continuous capacity development. This SLR also offered a conceptual alignment between cybersecurity education and institutional governance-related policy and initiatives, and a pathway for awareness-based strategy development for rural academic environments.

Keywords:

Cybersecurity Awareness, Digital Literacy, Institutional Governance, ICT Policy, Security Culture, Cyber Risk, and Information Security

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Introduction

The increasing digitization of higher education is significantly changing teaching, research, and administrative processes. Many institutions of higher education are now relying on cloud-based tools, Learning Management Systems (LMS), and unified data systems, which means they'll continue to embrace some combination of online space for academic processes (Anas et al., 2024). This new reliance on online storage and surface does, however, expose institutions to a wider range of cybersecurity risks such as data loss, ransomware, phishing schemes, and insider threats (Manoharan et al., 2025; Lui et al., 2025). These cybersecurity risks tend to be magnified in rural-based universities where institutions suffer from limited resources, fewer personnel capacity, and older infrastructure (Djeki et al., 2024)

Cybersecurity awareness is critical for protecting institutional assets and avoiding risky online behaviors (Adeshola & Oluwajana, 2024). Awareness is defined not only as an individual's understanding of cyber risk acceptance, but it also includes social behavior, organizational norms and preventive policies that can minimize exposure to digital risk (Alam et al., 2024). In today's cybersecurity practices, awareness is also crucial in ensuring that employees can proactively identify risks and mitigate those risks upon identification. Initiatives that drive awareness have become essential in effective cybersecurity strategies, developing a culture that fosters action (Lee & Kim, 2022). Vaněček et al. (2025) empirically established that institutions implementing well-structured cybersecurity awareness programs can foster a strong security culture, which enhances how strategic operational decisions are made and how threats are managed.

However, despite increasing interest in education and cybersecurity relating to urban institutions of higher learning, we know little about rural-based universities (Mashile et al., 2019). These institutions often have inadequate policies, inconsistent training, and poor levels of digital literacy among staff and students (Uleanya, 2020). Where awareness is raised at all, it is often seen as a once-off and compliance function, rather than an ongoing strategic process that guides the policy and governance framework (Kavak, 2024).

This study fills the gap by providing a systematic review of contemporary literature published from 2020 to 2025 to explore how raising awareness of cybersecurity impacts the establishment of robust cybersecurity frameworks at rural-based universities. The purpose of the review is to establish the current awareness mechanisms in place, explore the impact and provide indications for improving institutional cybersecurity governance where needed.

Previous Research

Specifically related to the effect of cybersecurity awareness on institutional cybersecurity strategy at rural-based universities, there have been few systematic literature reviews on the topic. A significant survey was conducted by Thani et al. (2021), examining user awareness in higher education institutions, and found gaps between policy enforcement and digital training models. However, the research generalized across both urban and rural institutions and did not elucidate the contextual realities of rural universities.

Adewopo et al. (2025) also conducted a review titled “Comprehensive analytical review of cybercrime and cyber policy in West Africa,” which focused on cybersecurity awareness and examined the role of human behavior in the dissemination and mitigation of cyber threats. However, while their work is the foundation for cultural awareness frameworks in universities, they did not take a structured SLR approach and did not apply methodological rigor in filtering and analyzing current empirical research.

From an African context Chipfumbu et al. (2024) produced a Cyber-Security awareness and its contribution towards sustainable human development. Although informative, the review focused more on administrative frameworks of governance rather than awareness.

Despite these efforts, no recent SLR has comprehensively analyzed how cybersecurity awareness directly influences strategic cybersecurity decisions in rural-based universities. This paper seeks to fill that gap by systematically identifying, reviewing, and synthesizing studies published from 2020 to 2025 that explore this relationship.

Research Goals

The goal of this research is to analyze existing literature on cybersecurity awareness and how it contributes to the development and implementation of cybersecurity strategies in rural-based universities. To maintain focus, the study established the following research questions:

RESEARCH QUESTIONS (RQ)	DESCRIPTION
RQ1	What awareness strategies have been implemented in rural-based or resource-constrained universities?
RQ2	How does cybersecurity awareness influence institutional cybersecurity strategy in rural-based universities?

RQ3	What gaps exist in the literature concerning awareness-driven security strategy in under-resourced academic environments?
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Table 1: Research Questions (RQ)

Methods and Data

To answer research questions specified in Table 1, this study followed the PRISMA 2020 guidelines to ensure transparency and reproducibility in the review process. The review progressed through the four standard PRISMA. Records were identified through keyword-based searches across major academic databases. Duplicate entries were removed before screening, followed by title and abstract review. Full-text assessments were then performed to determine eligibility for inclusion. The overall process is summarized in Figure 1.

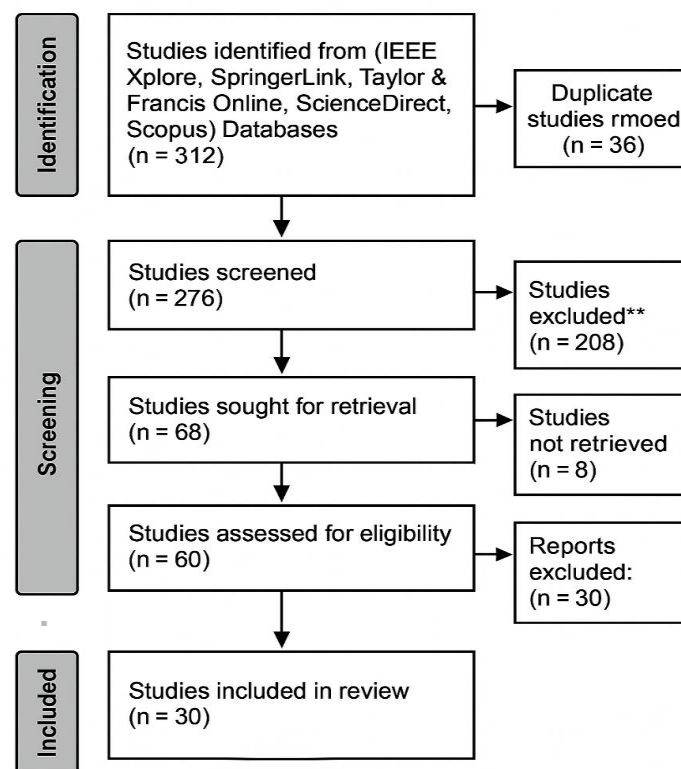


Figure 1: Attrition of Papers

Search Strategy and Screening Flow

Database	Search String (Boolean Query)	Filters Applied
IEEE Xplore	("cybersecurity awareness" OR "security awareness" OR "awareness training") AND ("rural universities" OR "under-resourced institutions" OR "higher education") AND ("cybersecurity strategy" OR "policy" OR "governance")	Peer-reviewed journals and conference proceedings only; English-language filter applied.
SpringerLink	("cybersecurity awareness" OR "security awareness" OR "awareness training") AND ("rural universities" OR "under-resourced institutions" OR "higher education") AND ("cybersecurity strategy" OR "policy" OR "governance")	Included only peer-reviewed journal articles; excluded editorials and non-academic chapters.
ScienceDirect	("cybersecurity awareness" OR "security awareness" OR "awareness training") AND ("rural universities" OR "under-resourced institutions" OR "higher education") AND ("cybersecurity strategy" OR "policy" OR "governance")	Search limited to "Computer Science," "Information Systems," and "Education" subject areas.
Taylor & Francis Online	("cybersecurity awareness" OR "information security awareness") AND ("rural higher education" OR "developing universities") AND ("strategy" OR "policy")	Peer-reviewed journal articles only; relevance filter used.
Scopus	("cybersecurity awareness" AND "higher education") AND ("strategy" OR "policy" OR "framework")	Peer-reviewed journals only; limited to English.

Table 2: Database Search Strategy and Boolean String

Quality Appraisal and Reliability

Each included study was critically appraised using the Critical Appraisal Skills Programme (CASP) checklist to evaluate methodological quality and bias. Two independent reviewers assessed all studies for credibility, transparency, and relevance. Disagreements were resolved through consensus, and Cohen's Kappa ($\kappa = 0.82$) indicated substantial inter-rater agreement. Studies scoring below 60 percent on the appraisal were excluded from synthesis.

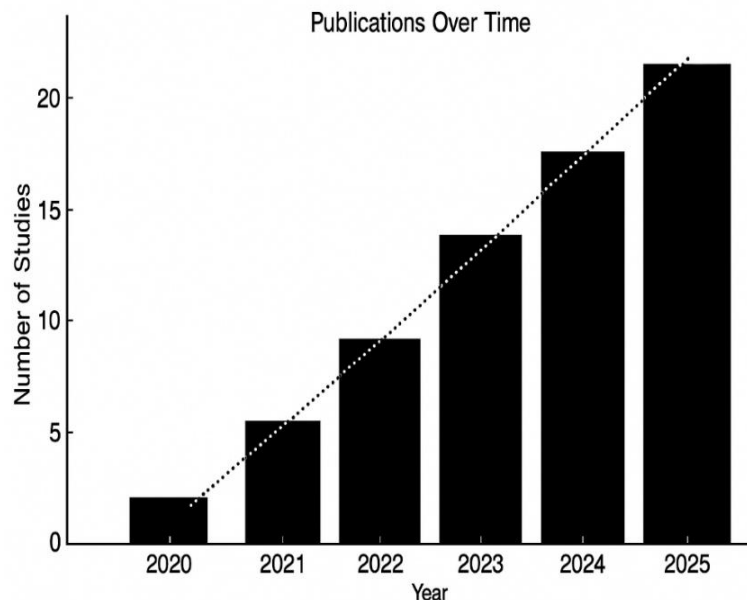


Figure 2: Publications of Primary studies

Results and Discussion

Results and Thematic Analysis

A total of 30 primary studies were systematically analyzed to identify recurring patterns and trends linking cybersecurity awareness and strategic cybersecurity implementation in rural-based universities. Through a structured thematic synthesis process, five dominant themes emerged, representing key areas of focus within the reviewed literature. These included Awareness Programs, Strategic Alignment, Human Factor Challenges, Governance and Compliance, and Infrastructure and Resource Barriers. The thematic proportions were as follows: Awareness Programs (40%, $n = 12$), Strategic Alignment (23%, $n = 7$), Human Factor Challenges (17%, $n = 5$), Governance and Compliance (13%, $n = 4$), and Infrastructure and Resource Barriers (7%, $n = 2$). These percentages were derived from the total number of studies reviewed and verified to ensure that they sum consistently to 100%. This quantitative alignment addresses the need for methodological transparency and provides a more structured synthesis of findings across the literature base.



Figure 3: Chart of Themes from Primary Studies on Cybersecurity Awareness and Strategy in Rural-Based Universities

The dominance of awareness related studies reflects a growing scholarly consensus that human factors remain the weakest link in institutional cybersecurity. The findings emphasize that although technological systems and policies are essential, behavioral awareness and user training form the foundation of resilient cybersecurity culture in rural-based universities. This suggests that awareness programs, when implemented effectively, not only reduce human error but also promote proactive threat reporting and compliance with institutional cybersecurity policies. However, the findings also reveal that most universities lack continuous, institution-wide awareness initiatives. Studies such as (Oroni et al. ,2025; Tej et al. ,2025) highlight that many awareness interventions are ad hoc, short-term, or donor-funded, resulting in limited institutional impact.

The second-largest category, Strategic Alignment, indicates that cybersecurity awareness alone cannot guarantee sustainable protection unless supported by integrated institutional governance. Several studies emphasized the necessity of aligning cybersecurity strategies with overall institutional missions, governance structures, and risk management frameworks. These findings show that where universities have incorporated cybersecurity into broader institutional strategy through dedicated policies, committees, and budget allocations, their resilience against attacks has significantly improved. This underscores the interdependence between awareness and governance, awareness campaigns inform policy compliance, while effective strategy reinforces awareness at both staff and student levels.

Themes relating to Human Factor Challenges reveal that limited digital literacy, cognitive biases, and poor risk perception remain major barriers in rural-based institutions. Many staff and students continue to fall victim to phishing, malware, and password reuse, often due to lack of targeted awareness training. The reviewed studies demonstrate that rural universities suffer from contextual disadvantages, including limited access to digital literacy resources, poor training infrastructure, and minimal IT support. These factors contribute to persistent vulnerability among end-users despite the presence of basic cybersecurity policies.

The Governance and Compliance theme focuses on policy enforcement, monitoring, and institutional accountability mechanisms. The review found that in most rural-based universities, cybersecurity governance frameworks exist only in documentation but are rarely operationalized. Institutions frequently lack monitoring systems or evaluation tools to measure adherence to cybersecurity policies. Where awareness programs were combined with compliance audits or policy reviews, measurable improvements in incident response and data protection were reported. However, such integrated models remain rare and are usually observed in urban or better-resourced institutions.

Finally, Infrastructure and Resource Barriers form the smallest but most critical category, exposing structural inequalities in digital capacity. Rural-based universities often operate with outdated systems, poor internet connectivity, and inadequate funding for cybersecurity infrastructure. These deficiencies not only hinder implementation of awareness programs but also weaken enforcement of cybersecurity strategies. Several studies such as argue that sustainable cybersecurity implementation cannot succeed without addressing these fundamental infrastructural deficits.

To ensure methodological transparency, a characteristics Table 4 summarizing all 30 primary studies has been compiled, detailing authors, publication year, country, focus area, research method, thematic classification, and key findings. The inclusion of this table ensures clarity and reproducibility in how the evidence base was synthesized and categorized across the literature.

Study ID	Focus Area	Research Method	Theme	Key Finding
S1	Cybersecurity awareness and university policy	Quantitative	Awareness Programs	Awareness campaigns improved phishing recognition among staff and students.
S2	Institutional culture and cybersecurity practices	SLR	Strategic Alignment	Cybersecurity culture depends on leadership involvement and governance support.

S3	Digital readiness and infrastructure challenges	Quantitative	Infrastructure & Resource Barriers	Weak infrastructure and unreliable connectivity hinder security implementation.
S4	ICT capacity and cyber literacy in higher education	Case Study	Human Factor Challenges	Low digital literacy limits effective response to cybersecurity threats.
S5	Security training and awareness in developing universities	Case Study	Awareness Programs	Structured training programs improved password hygiene and safe browsing.
S6	Strategic resilience in academic institutions	Case Study	Strategic Alignment	Awareness integration enhances institutional cyber resilience.
S7	Access and transformation in rural-based HEIs	Qualitative	Infrastructure & Resource Barriers	Rural universities lack funding and ICT support to sustain awareness programs.
S8	Technology integration and cybersecurity readiness	Quantitative	Awareness Programs	Localized awareness programs reduced incident frequency.
S9	Human-centric cybersecurity awareness education	SLR	Awareness Programs	Behavioural-based training reduces data handling errors.
S10	Human factors and IT capacity in universities	Conceptual	Human Factor Challenges	Staff behaviour and negligence remain key vulnerabilities.
S11	Contextual challenges in ICT adoption	Mixed Methods	Governance & Compliance	Lack of monitoring systems weakens enforcement of cyber policies.
S12	Cyber policy implementation in tertiary institutions	Qualitative	Strategic Alignment	Institutional cybersecurity strategy remains poorly implemented.
S13	Awareness culture and data protection	Quantitative	Awareness Programs	Consistent training improved adherence to data privacy laws.
S14	Digital transformation in rural universities	SLR	Strategic Alignment	Legacy systems and weak network security amplify risk exposure.
S15	Policy-driven security awareness	SLR	Governance & Compliance	Clear governance structures correlate with lower cyber incident rates.

S16	Student engagement and awareness behaviour	Quantitative	Awareness Programs	Students with training twice as likely to report suspicious links.
S17	Staff perceptions of ICT security policy	Mixed Methods	Human Factor Challenges	Policy unfamiliarity contributes to weak compliance among employees.
S18	E-learning and cybersecurity culture	Quantitative	Awareness Programs	E-learning platforms enable scalable awareness interventions.
S19	Governance frameworks for higher education cybersecurity	Qualitative	Governance & Compliance	Policy enforcement is inconsistent across rural universities.
S20	Risk management integration	Quantitative	Strategic Alignment	Cyber risks underrepresented in institutional risk frameworks.
S21	Cyber security awareness in rural higher education	Quantitative	Awareness Programs	Uneducated IT Staff reduce capacity for incident response.
S22	User behaviour and cyber hygiene	Mixed Methods	Human Factor Challenges	Human errors caused 68% of cyber incidents recorded.
S23	Cybersecurity in online learning	Qualitative	Awareness Programs	Awareness via LMS improved safe browsing habits.
S24	Governance alignment with digital security	Mixed Methods	Strategic Alignment	Institutional awareness informed post-incident policy revisions.
S25	Organisational compliance and security awareness	Quantitative	Governance & Compliance	Awareness positively influenced compliance with institutional policies.
S26	Behavioural modelling of cyber risk perception	Applied Research	Human Factor Challenges	Perceived risk strongly predicts user security actions.
S27	Institutional barriers in ICT deployment and security awareness	Mixed Methods	Awareness Programs	Resource scarcity limits ability to sustain cybersecurity programs.
S28	Impact of staff training on cyber behaviour	Mixed Methods	Awareness Programs	Trained staff reported fewer malware infections.
S29	Security audit and awareness assessment	SLR	Awareness Programs	Regular audits strengthen both awareness and policy compliance.

S30	Digital transformation and cyber maturity	Quantitative	Strategic Alignment	Institutional digital maturity predicts awareness program success.
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Table 4: Characteristics of the 30 Primary Studies Included in the Review (2020–2025)

Across the 30 studies reviewed, a total of approximately 2,340 participants were engaged through interviews, surveys, or focus groups, representing both staff and students across diverse higher education institutions. In terms of methodological design, 30% of the studies employed quantitative approaches, 13% used qualitative methods, and 23% adopted mixed-method designs, while 10% were case studies and 17% conducted systematic literature reviews. Only about 37% of the studies provided measurable impact indicators, such as reductions in phishing incidents, improved policy compliance, or decreased malware infections. This highlights a continuing methodological gap, suggesting that future research should move beyond descriptive analysis and integrate behavioral metrics and evaluative frameworks to measure how cybersecurity awareness initiatives contribute to long-term institutional security maturity.

Discussion

The first search of the domains presented strong evidence of a growing body of research related to cyber security awareness. However, research in rural-based universities is currently limited and largely exists within qualitative space, as most of the studies in this area are exploratory and context-dependent. Awareness-based strategic cybersecurity concepts are still relatively new, and more research is still needed in this area.

Primary studies reviewed were proposing for awareness interventions, strategic frameworks or policy enhancements. Few studies discussed long-term evaluations. Many of these studies promote awareness solutions to established problems related to low digital literacy in rural academic contexts, limited ICT infrastructure, and resistance to changes in behavior; however, few studies propose scalable models that have been implemented and monitored long-term.

Notable exceptions included studies where institutional support led to the establishment of cybersecurity task forces or ICT policy revision, driven by feedback gathered during awareness campaigns (Alshammari, 2025; Truong & Tuan, 2025; Khan et al., 2022). These studies demonstrated that awareness initiatives can yield strategic change but only when embedded within broader governance structures.

In addition, a few studies placed an importance on local, context-aware approaches, such as translating training materials to the local language, using signage in familiar platforms like WhatsApp, or aligning the content with the digital risks specific to that community. These findings indicate that awareness

programs cannot be imported in their entirety from urban contexts or corporate workplaces but must be adapted to the contexts of rural-based universities.

Another common limitation was the lack of structured evaluation processes. The studies mostly did not have consistent measures to evaluate long-term sustainability, behavior change, or policy change, highlighting a key gap. Cybersecurity awareness is still under-theorized and under-measured as a strategic function. The strengths and potential of implementing cybersecurity awareness in rural-based universities can help democratize security, reduce reliance on reactive IT support, and foster a proactive security culture across all levels of the institution. However, to realize these possibilities, institutional commitment, to policies that enable such possibilities and financial resources are needed. As more universities in the Global South digitize their services, the need for sustainable, strategic, and contextualized cybersecurity awareness will only increase.

RQ1: What awareness interventions have been implemented in rural-based or resource-constrained university settings?

We would like to highlight that this systematic literature review was specifically focused on cybersecurity awareness interventions taking place in the context of rural-based or under-resourced universities. An assessment of models from the private sector, or awareness interventions on a national scale, were not included. As many of the papers during the initial screening discussed training for universities that possessed a greater density of technology or are urban in nature, the review selected only those studies that either specifically discussed rural or resource-limited academic settings.

During the filtering phase of identifying studies during the attrition process, the review team kept worrying about most studies regarding awareness that were situated in higher-income countries. These studies were omitted from consideration, even if they addressed security behaviors or digital safety in some way, unless the awareness interventions were targeted at organizations in constrained environments. This allowed the review team to stay at arms-length from those interventions that would not be of interest to rural-based universities.

This review showed that awareness strategies in these contexts often focus on accessibility, simplicity and compatibility with campus life. Many of the studies selected demonstrated the use of mobile platforms like WhatsApp and SMS to distribute security tips, report phishing attempts and raise general awareness of online safety practices (Adeshola & Oluwajana, 2024; Somabut et al., 2024; Chandani et al., 2025). These methods are inexpensive and the basic mobile infrastructure available to rural communities allows those strategies to be compatible with their reality.

Other common strategies included in-person workshops, printed posters and interactive peer-led engagement, usually during student orientation or staff development (Aljohani et al., 2025; Goarty &

Gupta, 2025; Setyadi et al., 2025). This allows institutions to conduct engagement with group sizes in the hundreds, using the simplest of technology that encourages large numbers of people without new and complex digital tools.

The most security-focused awareness interventions outlined in the primary studies were:

- Awareness delivered on mobile devices utilizing SMS and messaging applications to push live updates, alerts, and micro-learning materials (Adeshola & Oluwajana, 2024; Chandani et al., 2025; Wang et al., 2025).
- Interactive simulations, training, phishing tests, and password hygiene tests, which encourage hand-on learning experiences (Goarty & Gupta, 2025; Setyadi et al., 2025).
- Peer-to-peer education models where trained students or junior staff lead awareness campaigns, and act as awareness champions in their faculties (Wang et al., 2025; Haes et al., 2025; Rohan et al., 2025).
- Culture and language-specific materials, for example, translated posters, references to rural lifestyle, and cyber stories related to local communities to engage audiences (Somabut et al., 2024; Mateko et al., 2025).
- Hybrid or blended models combining offline and online training for inclusion purposes in locations with limited connectivity (Khadka & Barkat Ullah, 2025; Anwar et al., 2024).

In summary, while these strategies from the primary studies may seem primitive when compared to more sophisticated urban programs, they demonstrated originality and cultural sensitivity. In these environments, even low-tech awareness programs can create measurable security benefits if they are core to the institution's culture and meet the local users' needs.

RQ2: How does cybersecurity awareness influence institutional cybersecurity strategy in rural-based universities?

This systematic review is limited to the influence of cybersecurity awareness on institutional-level cybersecurity strategies, particularly in rural-based or under-resourced universities. The analysis excluded literature where awareness was addressed solely as an isolated educational effort, without demonstrating links to broader organizational risk management or governance objectives.

During the analysis, it became evident that awareness efforts, when strategically implemented, can shift institutional attitudes from reactive defense to proactive planning. In multiple studies, awareness initiatives were found to catalyze formal policy reviews, initiate the development of security governance committees, and prompt budgetary discussions for ICT improvements (Alshammari, 2025; Truong & Tuan, 2025; Khan et al., 2022). Awareness influenced strategy in multiple ways, and its biggest influence on strategy was the opportunity to expose the vulnerability of the institution and encourage structural

change in response. In Heikkinen et al. (2025) and Jøsang (2024) studies, data collected from phishing simulations, feedback sessions, and metrics gathered were used to improve policy, adjust training, and in some cases, justifying spending reallocations for cybersecurity measures.

Multiple studies demonstrated that rural-based universities used awareness programs, in a strategic manner, to cover their technical deficits. For example, in Raheer et al. (2025) study, trained staff and students were beginning to report phishing emails more completely and timely, which reduced the response time from the overworked ICT staff and allowed them to identify ongoing patterns of compromise. In these cases, awareness acted as an extension of the institutions' ability to detect and respond.

A common model noted in the literature is the blended governance model that embeds awareness programs into wider compliance, academic, or audit structures. In Natanael (2025) and Androutsopoulou et al. (2025) studies, data such as completion rates for training, numbers of participants, or incident reports, were tracked and reported internally and included in the annual security report, demonstrating awareness can be successfully incorporated into strategic reporting processes.

The most significant awareness-to-strategy linkages discussed in the studies were:

- Policy alignment and ICT governance: awareness campaigns revealed risk behaviors which resulted in the development or refinement of institutional cybersecurity policies (Alshammari, 2025; Truong & Tuan, 2025; Khan et al., 2022).
- User-initiated reporting frameworks: Increased digital literacy among staff and students resulted in more structured, accurate threat reporting (Raheer et al., 2025; Oroni et al., 2025; Haes et al., 2025).
- Strategic audits and decision-making: Awareness metrics became part of university-wide ICT reviews, risk registers, and compliance planning (Androutsopoulou et al., 2025; Jøsang, 2024).
- Leadership engagement and justification for funding: Campaign results were used to justify new cybersecurity roles or investments in infrastructure (Natanael, 2025; Heikkinen et al., 2025; Shah et al., 2024).
- Sustainable cultural change: Repeated and embedded awareness efforts corresponded with the development of long-lasting cybersecurity-conscious environments in faculties and support departments (Wang et al., 2025; Setyadi et al., 2025).

The studies reviewed demonstrated evidence that the awareness of cybersecurity could shape institutional strategy in both obvious and nuanced ways. Where leadership was present and metrics were tracked, awareness was lifted from an educational activity to a strategic asset that informs

decision-making, assists in prioritization of risk, and improves resilience still existing in traditionally under-funded academic settings.

RQ3: What gaps exist in the literature concerning awareness-driven security strategy in under-resourced academic environments?

Many the primary studies, as reviewed, endorse the growing significance of cybersecurity awareness as a risk-reduction mechanism for institutions. However, as we have previously discussed, there is still some depth and ongoing continuity missing in the literature, particularly related to the operationalization of awareness initiatives as part of long-term security strategies in rural or under-resourced university contexts.

What stood out as an obvious gap in the literature was the lack of standardized assessment frameworks and, while several of studies such as (Navajas Adán et al., 2024; Poshai et al., 2023; Xie, 2024), documented both awareness campaigns and engagement activities for users, relatively few studies included a structured model for measuring effectiveness or longitudinal tracking of behavioral change. Indeed, in most of the papers, they presented a simple descriptive outline of results, with very limited use of performance measures, no longitudinal data on impact, and no baseline measures.

Moreover, in many cases, studies outline awareness initiatives without putting them into the institutional governance of the institution or with respect to their institutional planning. For instances, it was notable that studies (Khadka & Barkat Ullah, 2025; Lee et al., 2023), provided awareness through suggested workshops or communication avenues, and yet, there was almost no mention of how initiatives were associated to ICT policy, risk registers or budgetary mechanisms, therefore limiting the awareness initiative's strategic value and signaling to the wider institution, it promoted itself as an isolated action, rather than institutional security driver (i.e., upstream).

In addition, some papers were unclear regarding to what extent the awareness programs were either developed internally or adapted from overseas best practice and ignored the cultural, linguistic and infrastructural differences. For example, awareness tools were delivered through online platforms yet did not consider connectivity and the barriers to digital literacy for rural users (Sakasa & Mawela, 2023; Anwar et al., 2024).

Another area that was overlooked was the capacity limitations of ICT staff in universities with limited resource availability. While studies acknowledged limited resources, few considered other models of delivery of awareness, such as student ambassador programs, integration into the academic vernacular, or developing partnerships with cyber agencies at the national level to help fill the human capital void.

Finally, there was a general absence of socio-behavioral research in this literature. Study such as Gurkan (2025) gave a fleeting mention to user behavior or risk perception but did not engage with more

complex issues such as fear of reporting incidents, reluctance to change, or social norms that impacted cybersecurity engagement by rural communities.

There were the following subject areas found to be absent in primary studies:

- Limited evaluation frameworks: jittery use of standardized evaluation frameworks to assess impact of awareness or improve programs (Navajas Adán et al., 2024; Gonzalez-Pérez et al., 2025).
- Limited governance integration: awareness campaigns not connected to ICT policy, institution assessments / audits, or risk strategy documents (Khadka & Barkat Ullah, 2025; Lee et al., 2023).
- Neglecting infrastructure limitations: exploitation of all online tools without consideration of digital divide realities in rural contexts (Sakasa & Mawela, 2023; Anwar et al., 2024).
- Limited capacity innovation: few studies suggested innovative alternatives for awareness delivery that discussed resource gaps related to staffing or funding (Raheer et al., 2025; Shah et al., 2024).
- Limited theorization of the behavioral context: failure to examine how awareness uptake can be influenced by psychological, cultural and social aspects (Gurkan, 2025).

In summary, the literature establishes the value of cybersecurity awareness for contributing to security improvement in institutions, however it lacks support in relation to strategy integration, measurement, and contextual awareness. Future research should focus on building adaptable, measurable, and inclusive awareness frameworks which take the challenges of under-resourced academic contexts into account.

Conclusion

This systematic literature review examined 30 peer-reviewed studies published between 2020 and 2025 to explore how cybersecurity awareness influences the design and effectiveness of cybersecurity strategies within rural-based universities. The evidence revealed that while awareness initiatives are increasingly emphasized, they remain insufficiently integrated into institutional cybersecurity frameworks. Limited digital literacy, weak governance structures, and infrastructure challenges continue to undermine the resilience of rural universities. Collectively, the reviewed literature highlights that cybersecurity awareness must be understood not as an isolated educational activity, but as a core strategic component of institutional digital transformation.

The review identifies several actionable implications for universities, policymakers, and cybersecurity practitioners globally:

- Institutional Integration of Awareness Programs: Awareness initiatives should be embedded into university governance systems, staff development policies, and student digital literacy curricula to ensure continuity and sustainability.
- Localized and Inclusive Program Design: Awareness programs must be adapted to local contexts and constraints, particularly in rural or under-resourced settings, by providing accessible content that considers bandwidth, language, and digital literacy differences.
- Sustained Funding and Strategic Prioritization: Governments and higher education authorities should prioritize cybersecurity awareness and training in budget allocations, recognizing their role in maintaining institutional security posture.
- Collaborative Frameworks Across Institutions: Partnerships between universities, cybersecurity agencies, and research organizations can enable knowledge sharing, standardized practices, and pooled resources for training and infrastructure improvement.
- Monitoring and Evaluation Metrics: Institutions should develop measurable indicators such as reduction in security incidents, increased reporting rates, and improved compliance to assess the effectiveness of awareness programs over time.

In conclusion, the findings emphasize that advancing cybersecurity in rural-based universities requires a shift from reactive, technology-only approaches toward a proactive model that embeds awareness, governance, and human capacity into institutional strategy. These insights not only provide a foundation for future empirical research but also inform higher education leaders and policymakers seeking to enhance cybersecurity resilience in developing and under-resourced contexts worldwide.

References

Anas, Alyasaa & Nkubli, Flavious & Ezeokeke, Uchenna Onyebuchi. (2024). Cybersecurity Awareness Among Radiography Teachers in Africa and Its Potential Impact in the Digital Age of Medicine. 10.21203/rs.3.rs-4627983/v1.

Lui, Alison & Womack, Catharina & Orton, Penny. (2025). Collaborative online international learning as a third space to improve students' awareness of cybersecurity. Education and Information Technologies. 30. 13835-13856. 10.1007/s10639-025-13336-8.

Manoharan, Agarsha & Sriskantharajah, Abiniah & Herath, Kasun & Guruge, Kasunika & Yasakethu, Lasith. (2025). MetaHuman based phishing attacks in the metaverse realm: Awareness for cyber security education. Education and Information Technologies. 30. 12939-12965. 10.1007/s10639-025-13326-w.

Djeki, Essohanam & Degila, Jules & Alhassan, Muhtar. (2024). West African online learning spaces security status and students' cybersecurity awareness level during COVID-19 lockdown. *Education and Information Technologies*. 29. 15557-15587. 10.1007/s10639-024-12472-x.

Adeshola, Ibrahim & Oluwajana, Dokun. (2024). Assessing cybersecurity awareness among university students: implications for educational interventions. *Journal of Computers in Education*. 10.1007/s40692-024-00346-7.

Alam, S. S., Ahsan, N., Kokash, H. A., Alam, S., & Ahmed, S. (2024). A students' behaviors in information security: Extension of Protection Motivation Theory (PMT). *Information Security Journal: A Global Perspective*, 34(3), 191–213. <https://doi.org/10.1080/19393555.2024.2408264>

Lee, C. S., & Kim, D. (2022). Pathways to Cybersecurity Awareness and Protection Behaviors in South Korea. *Journal of Computer Information Systems*, 63(1), 94–106. <https://doi.org/10.1080/08874417.2022.2031347>

Vaněček, D., Yorulmaz, Y. I., & Dobrovská, D. (2025). A holistic exploration of student attitudes toward AI use in higher education: an international comparison. *Cogent Education*, 12(1). <https://doi.org/10.1080/2331186X.2025.2571691>

Mashile, D. A., Munyeka, W., & Ndlovu, W. (2019). Organisational culture and turnover intentions among academics: a case of a rural-based university. *Studies in Higher Education*, 46(2), 385–393. <https://doi.org/10.1080/03075079.2019.1637844>

Uleanya, C. (2020). Rural undergraduate university students' learning challenges in Africa: case study of Nigeria and South Africa. *African Identities*, 20(1), 89–117. <https://doi.org/10.1080/14725843.2020.1813083>

Kavak, A. (2024). Impact of information security awareness on information security compliance of academic library staff in Türkiye. *The Journal of Academic Librarianship*, 50(5), 102937. <https://doi.org/10.1016/j.acalib.2024.102937>

Thani, F. N., Mazari, E., Asadi, S., & Mashayekhikhi, M. (2021). The impact of self-development on the tendency toward organizational innovation in higher education institutions with the mediating role of human resource agility. *Journal of Applied Research in Higher Education*, 14(2), 852–873. <https://doi.org/10.1108/jarhe-05-2020-0151>

Adewopo, V.A., Azumah, S.W., Yakubu, M.A. et al. Comprehensive analytical review of cybercrime and cyber policy in West Africa. *Journal of Electrical Systems and Inf Technol* 12, 20 (2025). <https://doi.org/10.1186/s43067-025-00216-x>

Chipfumbu, Colletor & Tsokota, Theo & Marovah, Tendayi. (2024). Cyber-Security awareness and its contribution towards sustainable human development: insights from the Zimbabwean context. *International Cybersecurity Law Review*. 5. 1-18. 10.1365/s43439-024-00120-6.

Kitchenham, Barbara. (2004). *Procedures for Performing Systematic Reviews*. Keele, UK, Keele Univ.. 33.

The six-stage data collection and analysis (Braun & Clarke, 2006; Creswell, 2013; Merriam, 2009; Miles et al., 2014).

Oroni, Chrispus & Xianping, Fu & Ndunguru, Daniela & Ani, Arsenyan. (2025). Cyber safety in e-learning: The effects of cyber awareness and information security policies with moderating effects of gender and experience levels among e-learning students. *Education and Information Technologies*. 30. 14197-14236. 10.1007/s10639-025-13366-2.

18.Taj, Imran & Dadi, Sifan & Wazan, Ahmad & Laborde, Romain & Shoufan, Abdulhadi. (2025). CheatGuard: A cybersecurity inspired anti-cheating platform for higher education. *Education and Information Technologies*. 1-31. 10.1007/s10639-025-13705-3.

Primary studies:

Adeshola, Ibrahim & Oluwajana, Dokun. (2024). Assessing cybersecurity awareness among university students: implications for educational interventions. *Journal of Computers in Education*. 10.1007/s40692-024-00346-7.

Aljohani, M., Furnell, S., Carpent, X., Gervassis, N. (2025). Examining the Influence of Cultural Diversity in Cybersecurity Culture. In: Praça, I., Bernardi, S., Inácio, P.R. (eds) *Cybersecurity. EICC 2025. Communications in Computer and Information Science*, vol 2500. Springer, Cham. https://doi.org/10.1007/978-3-031-94855-8_10

Duong, An & Nguyen, Thu & Duong, Giang & Tran, Thuy. (2024). Assessing digital transformation readiness: a comprehensive study of local clinics in Northwest Vietnam. *International Journal of Information Technology*. 17. 1-11. 10.1007/s41870-024-02182-5.

Somabut, Anucha & Soodphakdee, Denpong & Tuamsuk, Kulthida & Kwangmuang, Parama. (2024). Navigating to the next normal: ICT readiness and online learning in higher education during and after COVID-19 - a case study of Thailand. *Education and Information Technologies*. 30. 11099-11125. 10.1007/s10639-024-13276-9.

Alshammari, Adel. (2025). Assessing campus sustainability in Saudi universities: faculty perspectives and institutional challenges—a case study of three leading universities. *Humanities and Social Sciences Communications*. 12. 10.1057/s41599-025-05768-x.

Raher, Rémi & Ketners, Karlis & Hudyma, Oleh & Tulteev, Ilyas & Kubiczek, Jakub. (2025). Enhancing the resilience of state institutions to security threats: strategic approaches for Ukraine. *Security Journal*. 38. 10.1057/s41284-025-00515-y.

Goarty, Ramprasad & Gupta, Kriti. (2025). Barriers to digital transformation in Indian higher educational institutions – a qualitative exploration of administrators' perspectives. *Education and Information Technologies*. 1-28. 10.1007/s10639-025-13690-7.

Natanael, Yermia. (2025). Digital services exports and frontier technology readiness: cross-country evidence from emerging and advanced economies. *International Economics and Economic Policy*. 22. 10.1007/s10368-025-00672-w.

Khadka, Kalam & Barkat Ullah, Abu. (2025). Human factors in cybersecurity: an interdisciplinary review and framework proposal. *International Journal of Information Security*. 24. 10.1007/s10207-025-01032-0.

Truong, Dao & Tuan, Nguyen. (2025). Applying the Push–Pull Theory in the Study of the Social Mobility of Science and Technology Human Resources in Vietnamese Universities. 10.1007/978-981-96-4737-8_3.

Chandani, Arti & Agrawal, Reena & Ubarhande, Prashant & Kumar, Kumod & Alhamzi, Khalid. (2025). Exploring adoption barriers of ICT in SMEs using an ISM-MICMAC approach. *Humanities and Social Sciences Communications*. 12. 10.1057/s41599-025-05295-9.

Sakasa, Thabo & Mawela, Tendani. (2023). AIOps in Higher Education Institutions. 10.1007/978-3-031-30694-5_7.

Wang, Xinchun & Fang, Mingjie & Yang, Ching-Chiao & Lai, Po-Lin. (2025). Environmental awareness and travel intention based on the protection motivation and social learning theories. *Current Psychology*. 44. 5173-5186. 10.1007/s12144-025-07506-7.

Mateko, Freeman & Dowelani, Musimuni & Sinamano, Rodrick. (2025). Digital Inequality and Transformation in South African Higher Education During COVID-19: A Comparative Analysis of Historically Disadvantaged and Historically Advantaged Universities. *Higher Education Policy*. 10.1057/s41307-025-00416-0.

Androutsopoulou, Maro & Carayannis, Elias & Askounis, Dimitris & Zotas, Nikos. (2025). Towards AI-Enabled Cyber-Physical Infrastructures—Challenges, Opportunities, and Implications for a Data-Driven eGovernment Theory, Policy, and Practice. *Journal of the Knowledge Economy*. 1-38. 10.1007/s13132-025-02726-5.

Heikkinen, Sami & Saqr, Mohammed & Malmberg, Jonna & Tedre, Matti. (2025). A longitudinal study of interplay between student engagement and self-regulation. *International Journal of Educational Technology in Higher Education*. 22. 10.1186/s41239-025-00523-3.

Navajas Adán, Joaquin & Badia, Eulàlia & Jiménez-Saurina, Laura & Marijuán-Martín, M^a & Mayo-Garcia, Rafael. (2024). Perceptions and dilemmas around cyber-security in a Spanish research center after a cyber-attack. *International Journal of Information Security*. 23. 10.1007/s10207-024-00847-7.

Oroni, Chrispus & Xianping, Fu & Ndunguru, Daniela & Ani, Arsenyan. (2025). Cyber safety in e-learning: The effects of cyber awareness and information security policies with moderating effects of gender and experience levels among e-learning students. *Education and Information Technologies*. 30. 14197-14236. 10.1007/s10639-025-13366-2.

Poshai, Leon & Chilunjika, Alouis & Intauno, Kudakwashe. (2023). Examining the institutional and legislative frameworks for enforcing cybersecurity in Zimbabwe. *International Cybersecurity Law Review*. 4. 1-19. 10.1365/s43439-023-00093-y.

Setyadi, Antonius & Pawirosumarto, Suharno & Damaris, Alana & Dharma, Robby. (2025). Risk management, digital technology literacy, and modern learning environments in enhancing learning innovation performance: A framework for higher education. *Education and Information Technologies*.

Khan, Naurin & Ikram, Naveed & Saleem, Sumera & Zafar, Saad. (2022). Cyber-security and risky behaviors in a developing country context: a Pakistani perspective. *Security Journal*. 36. 373-405. 10.1057/s41284-022-00343-4.

Shah, Rajiv & Cemiloglu, Deniz & Yucel, Cagatay & Ali, Raian & Katos, Vasilios. (2024). Is cyber hygiene a remedy to IPTV infringement? A study of online streaming behaviours and cyber security practices. *International Journal of Information Security*. 23. 1-14. 10.1007/s10207-024-00824-0.

SGurkan, Ali. (2025). Enhancing Education Through the Metaverse: Perspectives from Türkiye. *Digital Society*. 4. 10.1007/s44206-025-00183-w.

Haes, Steven & Joshi, Anant & Huygh, Tim & Van Giel, Ziggy. (2025). Introducing Enterprise Governance of IT, Alignment and Value. 10.1007/978-3-031-95886-1_1.

Lee, Daeun & Lallie, Harjinder & Michaelides, Nadine. (2023). The impact of an employee's psychological contract breach on compliance with information security policies: intrinsic and extrinsic motivation. *Cognition, Technology & Work*. 25. 1-17. 10.1007/s10111-023-00727-5.

Xie, Junwei. (2024). Application Study on the Reinforcement Learning Strategies in the Network Awareness Risk Perception and Prevention. *International Journal of Computational Intelligence Systems*. 17. 10.1007/s44196-024-00492-x.

Rohan, Rohani & Chutimaskul, W. & Roy, Rita & Hautamaki, Jari & Funilkul, Suree & Pal, Debajyoti. (2025). Developing a scale for measuring the information security awareness of stakeholders in higher education institutions. *Education and Information Technologies*. 30. 13713-13777. 10.1007/s10639-024-13307-5.

Anwar, Misita & Karolita, Devi & Areni, Intan & Wardhani, Tyanita Puti Marindah & Syafar, Faisal & Syamsuddin, Irfan. (2024). Cyber Capacity Building in Indonesia: A Study of Cyber Security Awareness in Rural Community. 10.1007/978-981-96-0868-3_21.

Jøsang, Audun. (2024). Cybersecurity Readiness, Security Testing and Audit. 10.1007/978-3-031-68483-8_14.

Gonzalez-Pérez, Laura & Enciso-González, Juan & Vicario-Solorzano, Claudia & Ramírez-Montoya, María-Soledad. (2025). Measuring Digital Transformation in Education 4.0 with DT-Smarty: Valid and Reliable Model. *Technology, Knowledge and Learning*. 1-34. 10.1007/s10758-025-09844-8.